Migrating to Azure Local

Agenda

- **Objective:** Explore end-to-end VM migration to Azure Local by using Azure Migrate
- **Scope:** Migration lifecycle (discovery, assessment, replication, cutover), Hyper-V/VMware support, networking, Azure Arc, security, monitoring, automation, and troubleshooting.

Introduction to Migrating to Azure Local

- **Migration to Azure Local**: Overview of strategies and tools for moving VM workloads.
- **Supported Platforms**: Hyper-V, VMware, and other on-premises environments.
- **First-Party & Third-Party Options**: Microsoft and partner solutions.
- **Key Considerations**: Planning, prerequisites, and supported configurations.
- **End-to-End Process**: Discovery, replication, migration, and validation.

Migration Options Overview

- First-Party Migration:
  - **Azure Migrate**: Central hub for discovery, assessment, and migration.
  - **No additional cost** for built-in tools.
- Third-Party Migration:
  - **Carbonite**, **Commvault**, **Veeam**: Partner solutions for specialized needs.
- Selection Criteria:
  - **Current infrastructure** compatibility.
  - **Migration goals** and complexity.
  - **Support for automation and reporting**.

Azure Migrate

- Azure Migrate:
  - **Discovery**: Identifies on-premises VMs and dependencies.
  - **Assessment**: Evaluates readiness and sizing for Azure Local.
  - **Migration**: Orchestrates replication and cutover.
- Supported VM Types:
  - **Hyper-V VMs** (Preview).
  - **VMware VMs**.
- Arc-Enabled VMs:
  - All migrated VMs are **Azure Arc-enabled** by default.

Azure Migrate Architecture and Components

- Azure Layer:
  - **Azure Migrate Project**: Orchestrates discovery and replication.
  - **Storage Accounts & Key Vault**: Store metadata and replication state.
  - **Azure Resource Manager (ARM)**: Manages resources across Azure and Azure Local.
- On-Premises Layer:
  - **Source Servers**: VMware/Hyper-V hosts.
  - **Source Appliance**: Discovers and prepares VMs.
  - **Target Appliance**: Receives migrated VMs on Azure Local.
  - **Arc Resource Bridge & Failover Cluster Manager**: Orchestration and provisioning.

Azure Migrate Migration Phases

- **1. Prepare:**
  - Complete prerequisites.
  - Deploy and register Azure Local.
  - Create Azure Migrate project and storage account.
- **2. Discover:**
  - Configure source appliance.
  - Discover on-premises VMs.
- **3. Replicate:**
  - Configure target appliance.
  - Select and replicate VMs.
- **4. Migrate & Verify:**
  - Migrate VMs to Azure Local.
  - Validate boot and data integrity.
  - Decommission source VMs.

Supported Configurations and Requirements

- **Source Environments**:
  - Hyper-V on Windows Server 2012 R2–2025.
  - VMware vCenter/ESXi 6.5–8.0.
- **Appliance OS**:
  - Windows Server 2022 (source and target).
- **Target Environment**:
  - Azure Local 2311.2 or later.
- **Guest OS Support**:
  - https://learn.microsoft.com/en-us/windows-server/virtualization/hyper-v/supported-windows-guest-operating-systems-for-hyper-v-on-windows.
- **Geographies**:
  - Asia-Pacific, Europe, United States (metadata storage).

Roles, Provider Registration, and Project Considerations

- Roles:
  - Entra tenant: Application Developer.
  - Azure subscription: Contributor, User Access Administrator.
- Resource Provider Registration:
  - Microsoft.DataReplication must be registered.
- Project and Appliance Pairing:
  - One source and one target appliance per Azure Migrate project.
  - Separate projects for Hyper-V and VMware sources.

Source and Target Prerequisites

- **Source (Hyper-V/VMware):**
  - Sufficient resources for appliance VM (16 GB RAM, 80 GB disk, 8 vCPUs).
  - VMs must be highly available (for clustered hosts).
  - Disks must be attached to CSV (Hyper-V).
  - BitLocker and encrypted disks must be disabled.
  - No shared disks or Azure Connected Machine Agent.
- **Target (Azure Local):**
  - Azure Arc resource bridge deployed.
  - Logical network and custom storage path configured.

Azure Migrate Project Setup

- Create Project in Azure Portal:
  - Select subscription and resource group.
  - Name the project and choose geography.
- Discovery and Replication:
  - Start discovery from the Azure Migrate dashboard.
  - Configure source and target appliances.
  - Review discovered VMs and initiate replication.

Migration Process

- Discovery:
  - Install and configure source appliance.
  - Power on VMs and ensure integration services/tools are installed.
- Replication:
  - Generate project keys for source and target appliances.
  - Download and install appliances (.VHD, .OVA, or .zip).
  - Register appliances and validate connectivity.
- Migration:
  - Select VMs, configure compute and storage settings.
  - Start migration and monitor progress.
  - Complete migration and clean up resources.

Network Prerequisites: Ports

- Firewall Ports (Hyper-V source / Azure Local target)
  - 3389 (RDP): Appliance remote administration.
  - 44368 (Appliance Manager): Access via https://<appliance-ip-or-name>:44368.
  - 5985 / 5986 (WinRM): Appliance-to-host communication.
  - 445 (SMB): Source ↔ Target appliance data movement.
- Firewall Ports (VMware source specifics)
  - 443 (Azure services & vCenter): Orchestrates **replication/migration** and **vCenter** connectivity.
  - 902 (ESXi heartbeat/snapshots): Data from ESXi snapshots; heartbeat to vCenter.

Network Prerequisites: URLs & General Guidance

- Required URLs
  - **Azure Migrate appliance URL access**.
  - ***.siterecovery.azure.com** for orchestration.
- General Guidance
  - Confirm **bidirectional** flows where noted.
  - Avoid overlapping **snapshot schedules** during replication.

Source Hyper-V Requirements & Constraints

- Supported Hyper-V Hosts: Windows Server 2012 R2 → 2025.
- Discovery Scope
  - **Standalone host VMs** are discoverable/migratable.
  - **Clustered hosts**: Only **HA VMs** are eligible; **non-HA VMs** on clusters must be **made HA** first.
- Appliance Capacity (Source)
  - Create a **Windows Server 2022** VM: **16 GB RAM**, **80 GB disk**, **8 vCPU**.
- Disk Preconditions
  - **CSV-attached** disks required for migration.
  - **DAS/SMB shares** on source **not supported**.
- Guest Prep
  - **Bring disks online**; persist **drive letters** (SAN policy).
  - **Uninstall Azure Connected Machine Agent** if present.
  - **Disable BitLocker**; **decrypt encrypted volumes**; **no shared disks**.

Target Azure Local Requirements & Setup

- **Azure Local Version: 2311.2 or later** for target migrations.
- **Azure Arc Resource Bridge (ARB)**
  - **Must exist**; created during deployment; verify via portal.
- **Logical Network**
  - Create and select a logical network for migrated VMs.
- **Custom Storage Path**
  - Configure storage path on ARB; ensure capacity for replicated disks.
- **Connectivity**
  - Source ↔ Target must reach each other (same network or VPN).

Azure Migrate Project Rules & Geographies

- 1:1 Appliance Pairing
  - **One source appliance** + **one target appliance per project**.
  - **Separate projects** for **Hyper-V** vs **VMware**.
- Tenant & Subscription Alignment
  - Project must be in **same tenant** as **Azure Local**.
  - Register **Microsoft.DataReplication** RP on all involved subscriptions.
- Project Geography
  - **Metadata only** stored in selected geography/region (e.g., **Central US, West US2; North/West Europe; SE/East Asia**).
  - VM data remains **on-prem**; **no customer data** leaves allocated region.

Hyper-V: Source Appliance Creation & Discovery

- Generate Project Key
  - In **Azure Migrate → Inventory → Start discovery → Using appliance → For Azure Local**.
- Install Source Appliance
  - **Template (.VHD)** method, or **Script (.zip)** via **AzureMigrateInstaller.ps1**.
  - Appliance host: **Windows Server 2022, 16 GB RAM, 80 GB disk, 8 vCPU**.
- Configure & Register
  - Use **Appliance Manager**; optional **preconfigured Entra ID app**.
- Discover VMs
  - Ensure VMs are **powered on** with **Hyper-V integration services** installed; wait for **green checkmark**.

Hyper-V: Target Appliance Creation & Registration

- Generate Target Key
  - In Azure Migrate → Replicate → Deploy and configure target appliance → Generate key.
- Install Target Appliance
  - Download .VHD or .zip, validate SHA-256, deploy on Azure Local.
  - VM sizing: Windows Server 2022, 16 GB RAM, 80+ GB disk, 8 vCPU.
- Register & Validate
  - Use Azure Migrate Target Appliance Configuration Manager; sign-in, MFA; status Registered/Validated.
- Provide System Info
  - FQDN, domain, credentials for the Azure Local system; Configure until Successful..

Hyper-V: Start Replication Wizard Part 1

- Basics Tab
  - Select **subscription**, **resource group**, **Target system** (**Azure Local**).
  - Prereqs show **green checks** (e.g., **Arc Resource Bridge**).
- Target Appliance Tab
  - Verify **connected** status (green check).
- Virtual Machines Tab
  - **Discovered VM list**; select up to **10** per batch.
- Target Settings Tab
  - **Storage account** (metadata only): **Standard**, **Public network access enabled**.
  - **Logical network**: pre-created static/dynamic logical network; **Reload** if missing.
  - **Storage path**: pre-created on ARB; ensure **free capacity**; **Reload** if missing.

Hyper-V: Start Replication Wizard Part 2

- Compute & Disks Tabs
  - **Rename target VMs**, assign **OS disk**, set **vCPU/RAM**; choose **disks to replicate** (OS disk mandatory).
- Review + Start
  - Verify all values; **Replicate**; stay on page until artifacts finalize.

Hyper-V: Migrate & Verify; Complete Migration

- Migrate VMs
  - In **Replications**, choose **Migrate** (bulk or single VM).
  - Optionally **shutdown** source VMs to avoid data loss.
- Verify Post-Cutover
  - In **Azure Local → Virtual machines**, confirm **Running** state; verify **disks & NICs**; test via **VMConnect/apps**.
- Complete Migration (Cleanup)
  - In portal: **Complete migration**; removes **protected items**, **seed artifacts** (**seed.iso**, seed disks).
  - Frees storage; allows **re-protect** later if needed.

VMware: Source Appliance Creation & Discovery

- Generate Project Key
  - Azure Migrate → Inventory → Start discovery → Using appliance → For Azure Local → VMware.
- Install Source Appliance
  - .OVA deployment **or script (.zip)** on **Windows Server 2022** VM (**16 GB / 80 GB / 8 vCPU**).
  - Install **VMware VDDK** (**8.0.0/8.0.1/8.0.2**; avoid **7.0.x**).
- Register & Configure
  - Optional **preconfigured Entra ID app**; add **vCenter credentials**; add **discovery sources**.
  - **Disable agentless application discovery** (not supported for Azure Local migrations).
- Discover VMs
  - Ensure **VMware Tools** installed; VMs **powered on**; wait for **green checkmark**.

VMware: Replicate, Migrate & Verify; Complete Migration

- Start Replication
  - Use **Replicate** wizard: **Basics / Target appliance / Virtual machines / Target settings / Compute / Disks / Review + Start**.
  - **Storage account** is for **metadata only**; ensure **Public network access**; **Standard** tier.
  - Select **logical network** and **ARB storage path**; ensure capacity; **Reload** if not listed.
- Migrate & Verify
  - Trigger **Migrate** from **Replications**; optionally **shutdown** source VM.
  - Validate **Running state**, **disk & NIC config**, and application behavior in **Azure Local**.
- Complete Migration
  - Use **Complete migration** to remove protected items and leftover **seed artifacts**; free capacity; enable re-protection.

Maintaining Static IPs During Migration

- Two Environment Types
  - **SDN-enabled Azure Local** → static IPs preserved **automatically**.
  - **Non-SDN Azure Local** → static IP migration requires **scripts / packages**.
- Why IP Preservation Matters
  - Ensures **application continuity**.
  - Prevents breaking **DNS**, **application bindings**, **firewall rules**, **legacy integrations**.

Static IP Preservation in SDN-Enabled Azure Local

- Automatic Preservation
  - Static IPs retained when assigned to **static logical networks**.
- Key Considerations
  - Multi-NIC VMs: SDN assigns **default gateway to all NICs** → remove gateway on secondary NICs.
- SDN Documentation References
  - **SDN Integration** (Azure Arc).
  - **Static Logical Networks** configuration guidance.

Static IP Preservation in Non-SDN Azure Local (Windows)

- Windows Support
  - Requires **Static IP Migration Package (.zip)**.
- Included Scripts
  - **Prepare-MigratedVM.ps1**
  - **Initialize-StaticIPMigration.ps1**
  - **Set-StaticIPConfiguration.ps1**
  - **Utilities.psm1**
- Execution Flow
  - Capture IP configuration pre-migration.
  - Store config in **InterfaceConfigurations.json**.
  - Apply configuration on first boot in Azure Local.
- Pre-Migration Requirements
  - VM must remain **powered on**.
  - Admin account required for scheduled tasks.

Static IP Preservation in Non-SDN Azure Local (Linux)

- Linux Support
  - Requires **Static IP Migration Package (.zip)**.
- Included Scripts
  - **prepare_migrated_vm.sh**
  - **azure_local_ip_helper.py**
  - **static_ip_manager.sh**
- Execution Flow
  - Capture IP configuration pre-migration via the **./prepare_migrated_vm.sh -o StaticIPMigration** command
  - Store config in a localized guest JSON file (**network_state.json**)
  - Apply configuration on first boot via a persistent systemd startup service.
- Pre-Migration Requirements
  - VM must remain **powered on**.
  - Root or sudo privileges required to inject startup binaries
  - python3 package must be installed for configuration parsing

Preparing VMs for Static IP Migration

- Source VM Requirements
  - **VMware Tools** or **Hyper-V Integration Services** installed.
  - VM must stay **powered on** through replication.
  - Admin credentials required to install scripts.
- Logical Network Requirements
  - Static IPs must be **free** in target network pools.
  - Target logical network must match **subnet layout**.
- Common Failure
  - If IP is already assigned → **Migration fails with "address already in use."**

Monitoring Migrations with Azure Migrate

- Diagnostic Logs
  - Enable logging on the **vault** associated with the Azure Migrate project
  - Supported categories: **ProtectedItems**, **Jobs**, **Errors**, more.
- Destinations
  - **Log Analytics Workspace** (query & dashboards).
  - **Azure Storage Account** (archival).
- Why Enable Diagnostics
  - Troubleshooting replication cycles.
  - Audit compliance & historical analysis.
  - Track failures across appliances.

Log Analytics for Migration Visibility

- Example Kusto Query

```
ASRv2ProtectedItems
| where TimeGenerated > ago(30d)
```

- Common Queries
  - Identify failing replications.
  - Track planned failovers (migrations).
  - Correlate appliance warnings and errors.
- Recommended Dashboards
  - Migration readiness overview.
  - Replication health over time.

Enabling Guest Management on Migrated VMs

- Guest Agent Requirements
  - Azure Local VM must run **Windows Server 2016+** or supported Linux distribution.
  - **Public network connectivity** required.
- Generation-Specific Requirements
  - **Gen1 VMs must be powered off** before enabling guest agent.
  - Gen2 can be powered on or off.
- Activation Flow
  1. Attach **guest agent ISO**.
  2. Run **install script** (Windows or Linux).
  3. Enable guest management from **Azure CLI or portal**.

PowerShell Automation for Migration

- Supported Automation Areas
  - Retrieve **discovered servers**.
  - Initialize **replication infrastructure**.
  - Create **replication jobs**.
  - Update and modify **protected items**.
  - Trigger **migration (planned failover)**.
  - Complete migration and remove protection.
- Key Cmdlets

```
Get-AzMigrateDiscoveredServer
Initialize-AzMigrateLocalReplicationInfrastructure
New-AzMigrateLocalServerReplication
Set-AzMigrateLocalServerReplication
Start-AzMigrateLocalServerMigration
Remove-AzMigrateLocalServerReplication
```

Troubleshooting Migration Failures

- Frequent Root Causes
  - Appliance registration issues.
  - Insufficient storage or full CSV.
  - VMs not highly available (Hyper-V clusters).
  - Disk discovery failures (“value cannot be null”).
  - IP conflicts during static IP migration.
  - Snapshot failures (VMware / Hyper-V).
- Troubleshooting Tools
  - **Diagnostic logs**.
  - **Appliance logs** under ProgramData.
  - **Azure portal Job IDs & correlation IDs**.
  - **Manual commands** (e.g., Get-Service, snapshot operations).

Troubleshooting: Appliance Registration Failures

- Common Causes
  - Incorrect **FQDN resolution** from appliance.
  - Missing **WinRM configuration** on Azure Local nodes.
  - Not using a **preconfigured Entra ID app** where required.
- Fixes
  - Add target system to **hosts file** manually: IP FQDN
  - Run `Enable-PSRemoting -Force` on each target node.
  - Re-register using Appliance Manager.

Troubleshooting: Target System Info Cannot Be Modified

- ## Cause
  - Target system details stored in **CredStore** cannot be edited once set.
- ## Resolution (only before replication begins)
  - Delete:
    `C:\ProgramData\Microsoft Azure\CredStore\TargetClusterCredentials.json`
  - Re-open appliance manager and re-enter values.
- ## Important Warning
  - **Not allowed after replication begins** — appliance must be removed & recreated.

Troubleshooting: Replication Fails (Capacity, HA, CSV)

- Root Causes
  - **CSV volume full** or insufficient free space.
  - VM not **highly available** on Hyper-V cluster.
  - OS disk not attached to **Cluster Shared Volume**.
- Fixes
  - Expand CSV volume as needed.
  - Convert VM to **high availability**.
  - Verify disk attachment inside Failover Cluster Manager.

Troubleshooting: Snapshot & Checkpoint Failures

- **Symptoms**
  - Snapshot creation fails during replication.
  - Migration job displays “Failed to delete snapshot”.
- **Root Causes**
  - Hyper-V host issues retrieving VHD metadata.
  - VMware snapshot conflicts or locked VMDK files.
- **Resolution Steps**
  - Validate `Get-VM`, `Get-VHD`, and manual checkpoint operations.
  - Resolve storage or host errors before retrying replication.

Troubleshooting: "Value Cannot Be Null" Disk Errors

- Error Message
  - Value cannot be null. Parameter name: FetchingHyperVDiskPropertiesFailed
- Cause
  - Hyper-V host fails to return correct **disk metadata (MaxInternalSize, ParentPath)**.
- Fix
  - Run WMI query via:
    `Get-WmiObject -Class Msvm_ImageManagementService`
  - Validate output XML and repair disk chain if needed.

Troubleshooting: VMs Fail to Shut Down During Migration

- Scenario
  - Source VM fails to power off during planned failover.
- Root Cause
  - WMI shutdown component failure.
- Fix

```
$vm = Get-WmiObject -Namespace root\virtualization\v2 -Query "Select * From Msvm_ComputerSystem Where Name='VMID'"

$shutdown = Get-WmiObject -Namespace root\virtualization\v2 -Query "Associators of {$vm} Where AssocClass=Msvm_SystemDevice ResultClass=Msvm_ShutdownComponent"

$shutdown.InitiateShutdown("TRUE", "Need to shutdown")
```

Troubleshooting: Cleaning Up Failed Migrations

- If failure happens *before* VM creation
  - No manual cleanup needed.
  - Retry migration.
- If failure happens *after* VM creation
  - Delete VM in Azure Local.
  - Validate deletion in Hyper-V Manager.
  - Do **not** delete:
    - Migrated disk(s)
    - Seed disk(s)
    - Network interfaces
  - Azure Migrate reuses these on retry.

Post-Migration: Guest Agent Enablement Issues

- Gen1 VMs
  - Must be **powered off** before attaching guest-agent ISO.
- If ISO attach fails
  - Symptoms: VM unmanageable in portal; DVD attach errors.
- Fix
  - Stop VM via Azure CLI:
    ```
    az stack-hci-vm stop --name <vmName> --resource-group <rgName>
    ```
  - Re-run guest agent enable command.

Final Migration Validation & Cleanup Checklist

- **Confirm VM Boot Success**
  - Validate app startup, OS services, event logs.
- **Verify Network Functionality**
  - Check static IP assignment (if applicable).
  - Test cross-subnet communication.
- **Storage Validation**
  - Confirm correct disks attached.
  - Validate VHD/VMDK size alignment.
- **Cleanup**
  - Complete migration in portal.
  - Remove source VMs from Hyper-V / vCenter.
  - Decommission source & target appliances if no longer needed.

Key Migration Unsupported Scenarios & Constraints

- Not Supported
  - VMs with **encrypted disks** (BitLocker or others).
  - **Shared disks** or clustered shared volumes inside guest.
- Additional Constraints
  - Only **one source + one target appliance** per project.
  - Can’t migrate VMs using **DHCP reservations** expecting MAC preservation.

Post-Migration: DNS & Network Re-Registration

- DNS Considerations
  - Ensure updated **A-records** reflect migrated IP.
  - Remove stale DNS entries from source environment.
- Network Verification
  - Validate **NIC name changes** after migration.
  - Confirm **default gateway** assignments (especially multi-NIC).
  - Re-apply **custom firewall rules**.
- Application Dependencies
  - Test API endpoints, service bindings, SMB shares, LDAP paths.

Post-Migration: Storage & Disk Integrity Validation

- Disk Checks
  - Confirm OS & data disks attached as expected.
  - Compare **VHD/VMDK sizes** pre- vs post-migration.
  - Validate disk **online status** & drive letters (Windows).
- SAN / fstab Considerations
  - Windows: ensure **SAN policy = OnlineAll**.
  - Linux: confirm **persistent mount identifiers**.
- Common Symptoms
  - Offline disks after cutover.
  - Incorrect disk mapping due to non-CSV sources.

Migration Cleanup

- Clean Up Target Environment
  - Remove **appliance VMs** (source + target).
  - Remove stored **logs** and **seed disks**.
  - Validate no stale NICs remain in logical networks.
- Clean Up Azure Resources
  - Delete Azure Migrate project (optional).
  - Delete replication vault metadata.
- Hyper-V / VMware Source
  - Decommission migrated VMs.
  - Validate cluster health after removal.

Best Practices for Large-Scale VM Migrations

- Batch Migration Strategy
  - Replicate in **10-VM batches** as supported.
  - Stagger cutover windows by workload tier.
- Operational Readiness
  - Ensure **monitoring & diagnostics** enabled.
  - Perform baseline checks on appliances weekly.
- Risk Mitigation
  - Pre-cutover **snapshot / checkpoint** in source (VMware only).
  - Document **fallback procedure**.

Automation for Enterprise-Scale Cutovers

- **Azure PowerShell Automation Includes**
  - Bulk `Start-AzMigrateLocalServerMigration` calls.
  - Automated validation workflows.
  - Scheduled migration windows.
- **Pipeline Integration**
  - CI/CD orchestration for migration waves.
  - Integration with Azure DevOps or GitHub Actions.
- **Benefits**
  - Predictability, repeatability, reduced human error.

Common Operational Questions

- Can Windows Server 2008 R2 be migrated?
  - Yes, with SP1 and patch KB3138612 installed.
- Are Secure Boot VMs supported?
  - Yes.
- Can VMware and Hyper-V be migrated in same project?
  - No.

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